

Financial Engineering and Risk Management

Assignment 1

Prof. Süleyman Başak; Instructed by Prof. Darcy Pu

17 Sep 2026

Grading Guideline:

Due by **7 Oct 2026**. Show all work to receive full credit.

Only email your **typed solutions in pdf** to our Teaching Assistant Mengyue Shan, 2401111051@gsm.pku.edu.cn.

To promote sustainability, please do NOT print it.

Must write down **every group member's name**, otherwise they will receive ZERO. Group members' names cannot be added/dropped after each submission.

Late submissions will receive ZERO with no exception.

1. Geometric Brownian Motion – 5 points

A stock price follows a geometric Brownian motion process with current price \$50. Assume the (instantaneous) expected return for the stock is $\mu = 18\%$ p.a. and its volatility is $\sigma = 30\%$ p.a.

- What is the probability distribution for the stock price in two years?
- Calculate the mean and standard deviation for the stock price in two years.
- Determine the 95% confidence interval for the stock price in two years.

2. Mean Reverting Processes – 5 points

Suppose the current time is t and x_t is the yield to maturity with continuous compounding on a zero-coupon bond that pays off \$1 at time T . Assume that x follows the process

$$dx_t = a(x_0 - x_t)dt + sx_t dW_t$$

where a , x_0 , and s are positive constants and W is a Brownian motion. What is the process followed by the bond price?

3. Arithmetic Brownian Motion – 10 points

A company's cash position, measured in millions of dollars, follows an arithmetic Brownian motion with a drift rate of 0.1 per month and a variance rate of 0.16 per month. The initial cash position is 2.0.

- What are the probability distributions of the cash position after one month, six months, and one year?
- What are the probabilities of a negative cash position at the end of six months and one year?
- At what time in the future is the probability of a negative cash position greatest?

4. Stochastic Calculus – 5 points

Show that

$$\int_0^t W_s dW_s = \frac{1}{2}W_t^2 - \frac{1}{2}t$$

where W is a Brownian motion process. *Hint: Consider the dynamics of W^2 .*

5. Stochastic Calculus – 5 points

Show that

$$\int_0^t s dW_s = tW_t - \int_0^t W_s ds$$

where W is a Brownian motion process.

6. Simulations – 10 points

Use your favorite programming language to simulate Arithmetic Brownian Motion, Geometric Brownian Motion, Ornstein-Uhlenbeck, and square-root processes.

For each process, use the same set of parameters to simulate it 500 times and compute their mean and variance at $t = 1$. The parameters to be used are: $\kappa = 0.5, \mu = 0.7, \sigma = 0.2, X_0 = 0.1$. Let $t \in [0, 1]$ with intervals $\Delta t = 0.01$.